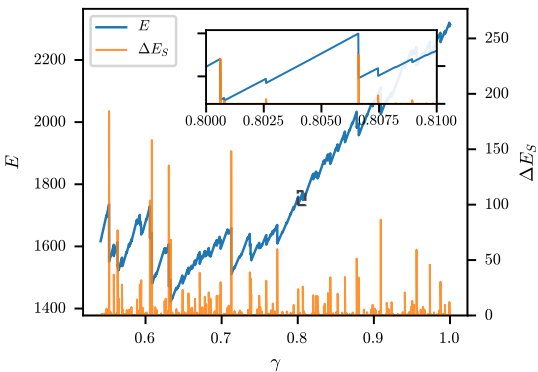
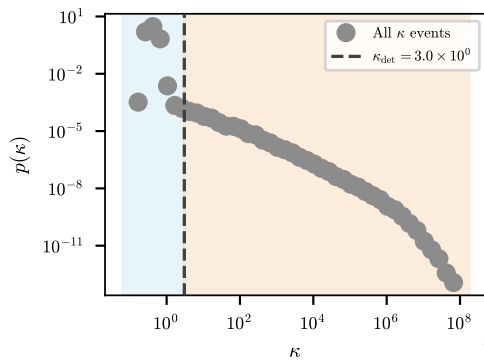
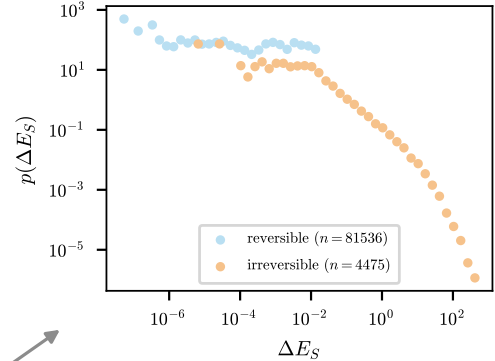


(a) Energy drops

(b) Drop detection: κ (c) Reversible / irreversible ΔE_S 

$$\Delta E_S = \widehat{E}_{n+1} - E_n + 1$$

$$\widehat{E}_{n+1} = E_n + V_0 \langle \sigma_{12} \rangle_n \Delta \gamma_n + \frac{V_0}{2} a_{1212, n} (\Delta \gamma_n)^2$$

$$D(\Delta E_{\min}) = \sup_{x \geq \Delta E_{\min}} |\widehat{P}_>(x) - P_{>}^{\text{TPL}}(x)|$$

$$\widehat{P}_>(x) = |\mathcal{X}_{\min}|^{-1} \sum_{\Delta E_S \in \mathcal{X}_{\min}} 1(\Delta E_S > x)$$

$$P_{>}^{\text{TPL}}(x) = \int_{\mathcal{X}_{\min}} p(u | \hat{\alpha}, \hat{\lambda}, \Delta E_{\min}) du$$

$$\mathcal{X}_{\min} = \{\Delta E_S : \Delta E_S \geq \Delta E_{\min}\}$$

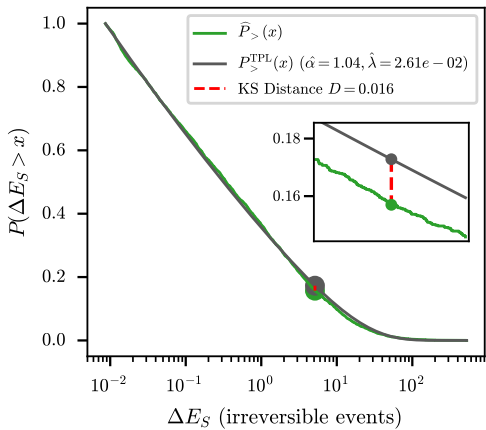
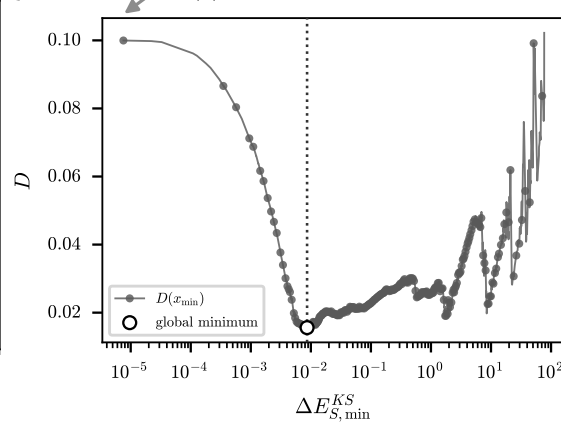
$$\Delta E_R < \Delta E_{R, \min} : \text{reversible}$$

$$\Delta E_R \geq \Delta E_{R, \min} : \text{irreversible}$$

$$p(\Delta E_S | \alpha, \lambda, \Delta E_{\min}) = \frac{\Delta E_S^{-\alpha} e^{-\lambda \Delta E_S}}{\mathcal{Z}(\alpha, \lambda, \Delta E_{\min})}$$

$$\ell(\alpha, \lambda) = \sum_{\Delta E_S \in \mathcal{X}_{\min}} \ln p(\Delta E_S | \alpha, \lambda, \Delta E_{\min})$$

$$(\hat{\alpha}, \hat{\lambda}) = \arg \max_{\alpha, \lambda} \ell(\alpha, \lambda)$$

(d) KS distance: irreversible ΔE_S (e) Global $\Delta E_{\min}$ 

(f) MLE fitting

